
ECON 212 — Principles of Microeconomics

Lecture Notes — Ch. 7: Market Efficiency

Fall 2026

Up until now, we've taken equilibrium prices as a given.

- When the government implemented price controls, we compared it to E.Q.
- When S or D shifted, we followed the E.Q.

But we never answered the question —

Q *Should E.Q. be the price?*

- *What does "should" even mean?*

To tackle this, we're going to follow this outline.

Outline

① Consumer Surplus

↔ How (in \$ terms) buyers benefit from participating in the market.

② Producer Surplus

↔ Mirror to ①: how do sellers benefit?

③ Maximizing Total Surplus

↔ How can we make ① + ② as large as possible?

★ Is this even the correct measure to use?

① Consumer Surplus

Start w/ Calvin & Hobbes comic

We make the same assumption that Calvin makes.

- Everyone has a "maximum value," or "reserve price," for the good in question. We call this the **willingness to pay**.
- + This shouldn't be a cause for concern. At a low price, you would be willing to buy a new Switch. At a high price you would not, so somewhere in between, you have to flip.

Imagine now that we are selling a good that each person wants exactly 1 copy of, say a Nintendo Switch. . .

- This is just for simplicity.

★ *Get one WTP for a Switch from a student.*

→ Ask them about what would happen if they were offered a Switch at \$100 less than this?

↪ \$200 less than?

↪ @ the same price?

↪ What if more?

Consumer Surplus

The difference between price and WTP is the **consumer surplus**.

→ *We now went from a hard-to-define notion of “goodness” to a quantifiable \$ value!*

Lining Everybody Up

Now suppose I have a room full of people, each with their own WTP.

★ *Collect WTP from several students.*

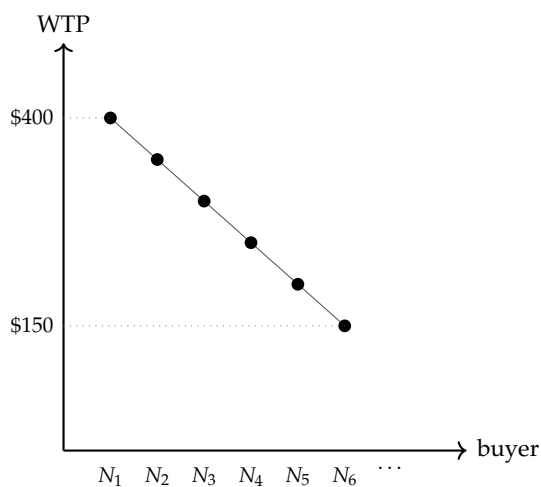
Now, let's do a simple thing and just line up people by their willingness to pay.

A worked set of numbers (added — the notes leave these blank)

Six buyers, sorted from highest WTP to lowest:

Buyer	N_1	N_2	N_3	N_4	N_5	N_6
WTP	\$400	\$350	\$300	\$250	\$200	\$150

Buyer N_n has $WTP_n = 450 - 50n$, so the “line drawn between them” is exactly $P = 450 - 50Q$.



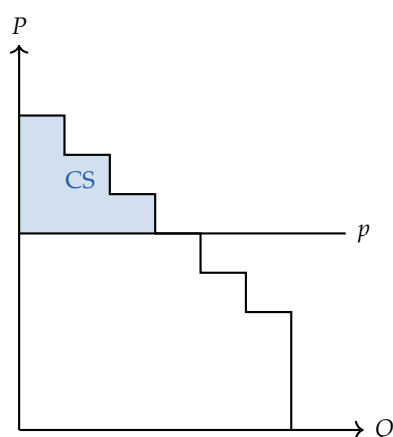
Next, draw a line between them...

Q *What have we here?*

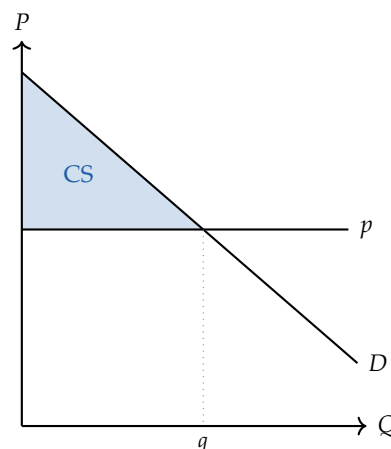
→ We call this just a **demand curve**.

Reading Consumer Surplus Off the Graph

Go through and add a price to that demand curve.



(a) six buyers
 $\$150 + \$100 + \$50 + \$0 = \$300$



(b) very many buyers
 $\frac{1}{2} \cdot 4 \cdot \$200 = \$400$

→ We can pick out consumer surplus just by looking at the graph!

Of course, when we have many buyers we would get a smooth-looking demand curve, and so CS is just the triangle.

A Few Notes/Questions

★ We'd assumed that each buyer only bought one unit. . .

Q What if they may want more?

→ Just label them "Brad-1" and "Brad-2" . . .

★ What is consumer surplus measuring?

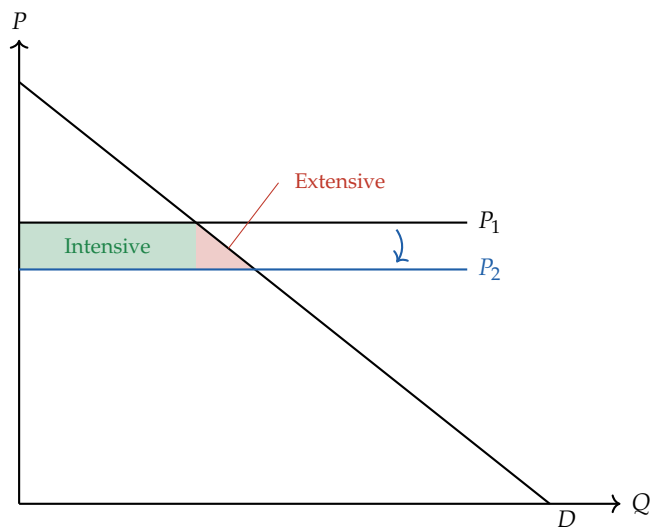
→ How valuable access to the market is for the buyers!

→ The WTP for the *right* to buy at a price.

→ Critically, this is what the buyers themselves value it at! We aren't imposing anything on them in terms of values.

★ *This is a truly beautiful element of Econ! We aren't imposing our values on others!*

★ What happens when the price is lowered?



Consumer surplus increases through two channels.

- ① **Intensive Margin:** buyers who were already buying now have more surplus.
- ② **Extensive Margin:** previously unengaged buyers now purchase the good and gain CS.

② Producer Surplus

This is “déjà vu” all over again. We’re going to do the same thing again, but this time w/ “cost” rather than “WTP.”

→ Implicit in both set-ups is that buyers & sellers are **price takers!** They aren’t haggling over the price.

Each seller has their own cost.

→ Make up a similar set of costs as the WTP from the first part.

★ Producer surplus is the difference b/w price and cost!

↪ How much \$\$\$ profit the seller makes.

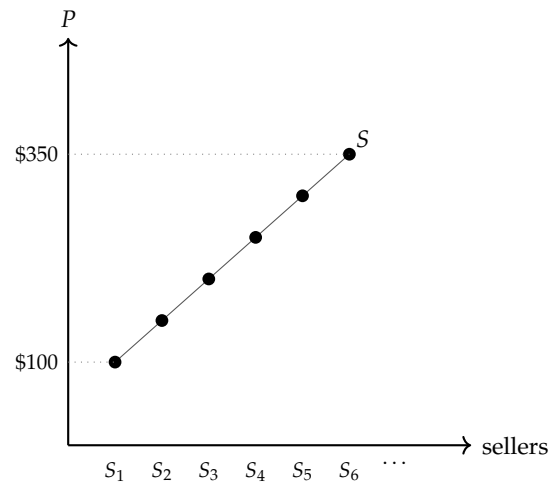
Of course, we can order them by their costs (lowest to highest this time) and get the supply curve.

The mirror-image numbers (added — the notes leave these blank)

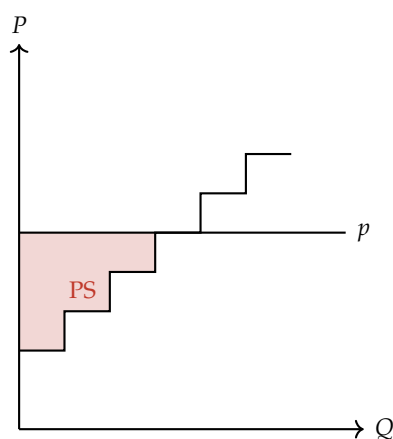
Six sellers, sorted from lowest cost to highest:

Seller	S_1	S_2	S_3	S_4	S_5	S_6
Cost	\$100	\$150	\$200	\$250	\$300	\$350

Seller S_n has $c_n = 50 + 50n$, so the line drawn between them is $P = 50 + 50Q$.

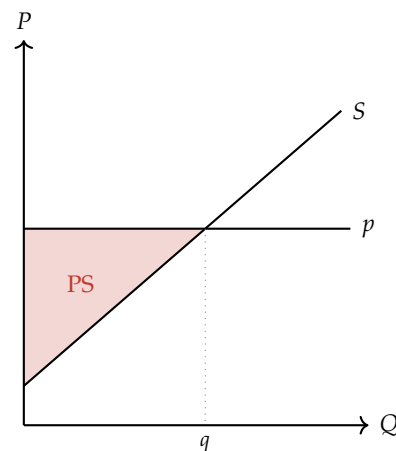


Add a price to the graph, and get out the producer surplus.



(a) six sellers

$$\$150 + \$100 + \$50 + \$0 = \$300$$



(b) very many sellers

$$\frac{1}{2} \cdot 4 \cdot \$200 = \$400$$

We can follow up w/ the same types of notes.

① What is producer surplus?

- ↪ The value to the sellers of being able to sell the good.
- ↪ Again, this is what the sellers themselves would at most pay.

② What happens to PS as the price rises?

- ↪ **Intensive Margin:** higher PS for those already selling.
- ↪ **Extensive Margin:** new PS for new sellers.

③ Maximizing Total Surplus

We now turn to the harder problem of making people as “well-off as possible”...

→ How should we “quantify” this?

↪ $CS + PS$.

→ We just add these together. Of course, you may have your own prejudices as to whose surplus matters more, but note the simplicity: we care about how much both sides would be willing to pay to make this market happen! In some sense, this is the “value” of the market.

Pretend now that you are (in a stretch of imagination) omniscient and (even more difficult) benevolent. How do you decide which exchanges to suggest, and what price to charge each buyer???

→ Should we just match on highest WTP + highest cost?

↪ That would maximize the # of transactions.

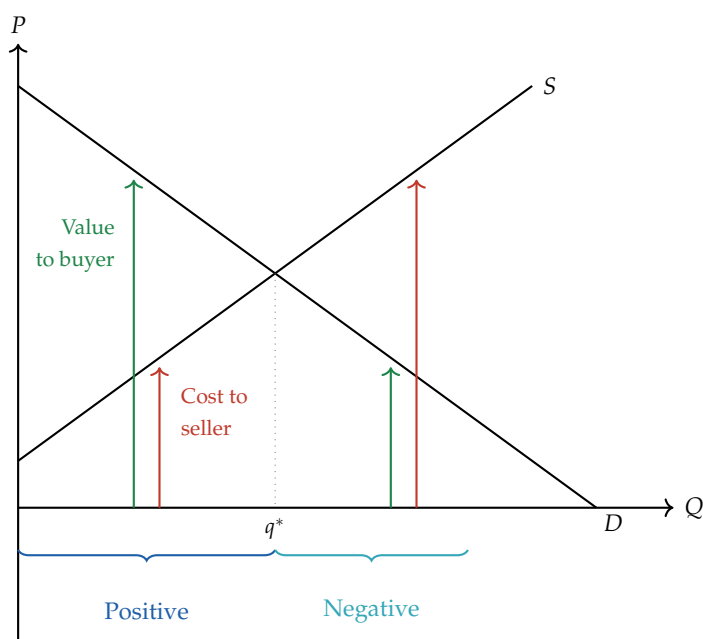
To do this, just examine total surplus.

Total Surplus

$$\begin{aligned}
 TS &= CS + PS \\
 &= (\text{value to buyers} - \text{price}) + (\text{price} - \text{cost to sellers}) \\
 &= \boxed{\text{value to buyers} - \text{cost to sellers}}
 \end{aligned}$$

So we just need to suggest exchanges that are positive in this respect.

→ Just look at the S & D graph again!



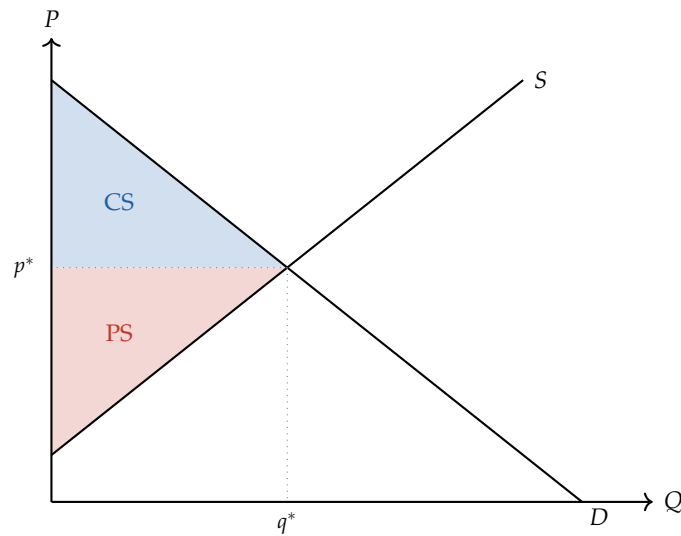
Where does the sign flip?

→ Right where we cross q^* !

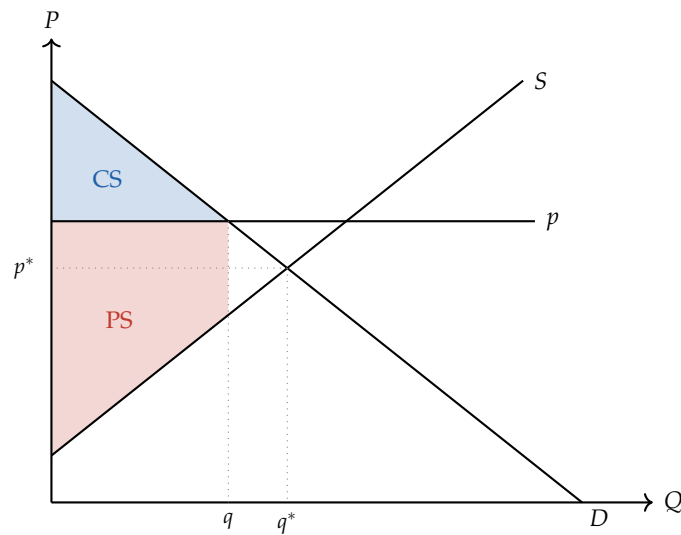
Is there a single price that facilitates this?

→ p^* .

So p^* is not only the market-clearing price, but also the surplus-maximizing one! Let's visualize this.



Q What do $CS + PS$ look like when $p > p^*$?



Q How are these benefits allocated?

- All consumers lose out.
- Some sellers gain on the intensive margin.
- Some sellers lose on the extensive margin.

By the numbers (added — the handwritten figures carry no numbers)

With $D : P = 450 - 50Q$ and $S : P = 50 + 50Q$, equilibrium is $p^* = \$250, q^* = 4$.

	CS	PS	TS
At $p^* = \$250$ (4 traded)	\$400	\$400	\$800
At $p = \$300$ (3 traded)	\$225	\$525	\$750
Change	−\$175	+\$125	−\$50

The \$50 that vanishes is the deadweight loss — $\frac{1}{2} \cdot 1 \cdot \100 . Note how it splits: sellers gain $3 \times \$50 = \150 on the intensive margin but lose \$25 on the extensive margin, netting +\$125. Consumers lose \$150 on the intensive margin and another \$25 on the extensive margin. The two \$25's are exactly the deadweight loss.

Efficiency

We say that equilibrium is efficient: it maximizes total surplus.

There are many reasons to think that this allocation is desirable.

Q List a few:

- It does the most “good.”
- It maximizes personal agency...
- The lowest-cost producers supply the good!
- The buyers w/ the highest WTP get the good!

★ Very important! ✓✓✓
→ An assumption we make.

We call all of these benefits the invisible hand.

But it isn't always hunky-dory...

Q What does our analysis leave to be desired?

→ **Equality**

+ Hard to define, but worthwhile nonetheless.

→ **Too much freedom...**

+ Sometimes your actions affect others. This is an externality.

→ **Not all sellers are price takers.**

+ Monopolists can set the price.

↪ They have market power.

Market
Failures

Together, these two are the market failures.

⇒ We are going to tackle externalities in the next section of the course, and market power in the following section.